

Preliminary Summary of Harmonic Readout and $c=1$ Area Sweep in an AB Two-Body Closed Phase System v4

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V4 implements the fermion-like recoil map as a direct A/B two-channel scattering matrix acting on the incident channels.

Abstract

This summary consolidates six preliminary experiments performed on an AB two-body closed phase system. V4 keeps the same scope and claims and incorporates a fermion-like recoil map into the same AB acceleration-readout frame as a direct A/B two-channel scattering action:

1. one-angle circumferential phase harmonic readout,
2. parameter sweep of the one-angle circumferential phase harmonic readout,
3. internal $c=1$ calibration and χ - τ area sweep,
4. parameter sweep of the internal $c=1$ calibration,
5. inverse-area compensation diagnosis in the χ - τ plane, and
6. extended sweep for native inverse-area readout, and
7. direct two-channel fermion-like recoil harmonic readout protocol.

The preceding ABC multigauge interference readout paper showed that, in a minimal ABC closed phase system that does not place a background space first, the measuring device itself can be defined as a complex phase wave inside the same system, and mass-like, momentum-like, and energy-like conserved readouts can be constructed from interference.

The present experiment group takes one step further. It verifies that an acceleration-like readout can be represented by harmonic readout in the AB two-body relation itself.

However, the AB two-body system alone could not determine whether the acceleration-like readout changes with the position phase difference, or distance, according to an inverse law or an inverse-square law.

The central achievement of this experiment group is not merely the display of harmonic motion. No external standard force, spring equation, gravitational equation, or Coulomb equation is introduced. No individual forces f_A and f_B are introduced. Using only the label-free two-body relation f_{AB} , an acceleration-like readout was constructed.

At the same time, the experiment clarified a boundary. In the AB two-body system, the measuring device is on the measured object itself. Therefore, although relative-distance change can be read, there is no independent gauge that can decide whether the change follows

L_{AB}

$1 / L_{AB}$

$1 / L_{AB}^2$

The gauge used to measure the interval is itself affected by the same closed phase system. Even after adding a temporal phase difference in addition to the spatial position phase difference, and even after imposing an internal $c=1$ -like calibration to reproduce a time-development-like effect, the conclusion did not change.

Thus, the determination of the distance exponent remains a task for a separate ABC three-body experiment with an independent metrical gauge.

Keywords: AB two-body closed phase system, label-free two-arc relative phase, harmonic readout, acceleration-like readout, $c=1$ internal calibration, χ - τ area, inverse-area diagnosis, native readout

1. Experiment List

No.	Experiment	Purpose	Main result
1	One-angle harmonic readout	Test whether D_{AB} , V_{AB} , and f_{AB} can be read label-free	Readable
2	One-angle parameter sweep	Robustness against initial deviation, period, and readout leakage	Harmonic oscillation detected in all cases
3	$c=1$ area sweep	Separate s from τ_{read} and test whether a $\chi\text{-}\tau$ plane stands	Area detected under independent τ
4	$c=1$ parameter sweep	Test whether $c=1$ is sufficient for area formation	Not sufficient
5	Inverse-area compensation diagnosis	Test whether $1/A_{\chi\text{-}\tau}$ remains natively on the closure-compensation side	Not detected
6	Native inverse-area extended sweep	Search broad conditions for $\alpha \approx 2$	Not detected natively
7	V4 fermion-like recoil protocol	Re-test acceleration-like readout by placing an A/B two-channel scattering matrix near the AB center	q_{out_factor} is not used as an operator; harmonic readout and the $\chi\text{-}\tau$ surface are re-read from the outgoing channel difference

2. One-Angle Circumferential Phase Harmonic Readout

2.1 Meaning of the Experiment

The AB pair was read as a closed phase system without placing an observer C.

The experiment did not introduce

f_A

f_B

as independent forces. It read only

f_{AB}

as relational compensation.

No standard gravitational equation, standard Coulomb equation, or standard spring equation was used.

2.2 Main Result

The main numerical results were:

Quantity	Value
case_count	32
observer_C_used	False
f_A_or_f_B_used	False
max_Q_closed_abs	0.0
oscillation_detected_all_cases	True
label_free_protocol_degenerate_all_cases	True
readout_decay_monotonic_all_cases	True
readout_off_decay_max_abs	6.49e-17
readout_strong_decay_min_abs	4.0004e-4

The result was:

Protocols F and B differ as internal representations,
but they are completely degenerate in D_AB and V_AB.

This means that reflection-type and pass-through-type protocols cannot be distinguished by label-free readout.

It was also confirmed that

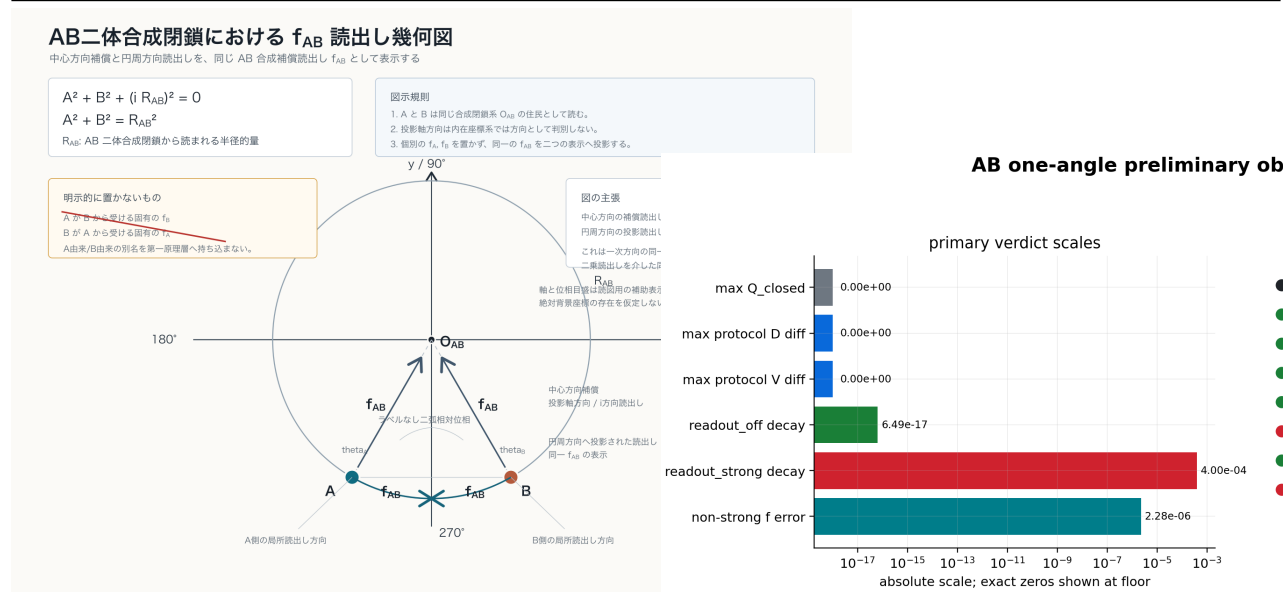
no decay occurs when readout is off,
and the envelope decays more strongly as the readout becomes stronger.

This is a falsification test showing that a readout wave can affect the envelope of the system when information is read out from the closed phase system.

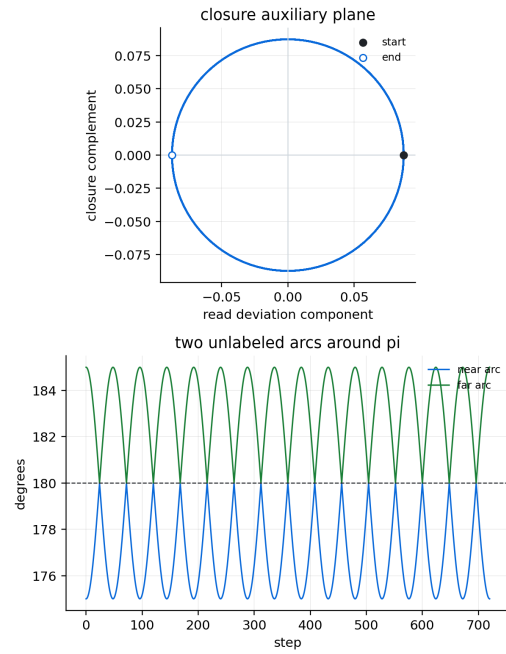
2.3 Figures

AB two-body geometry

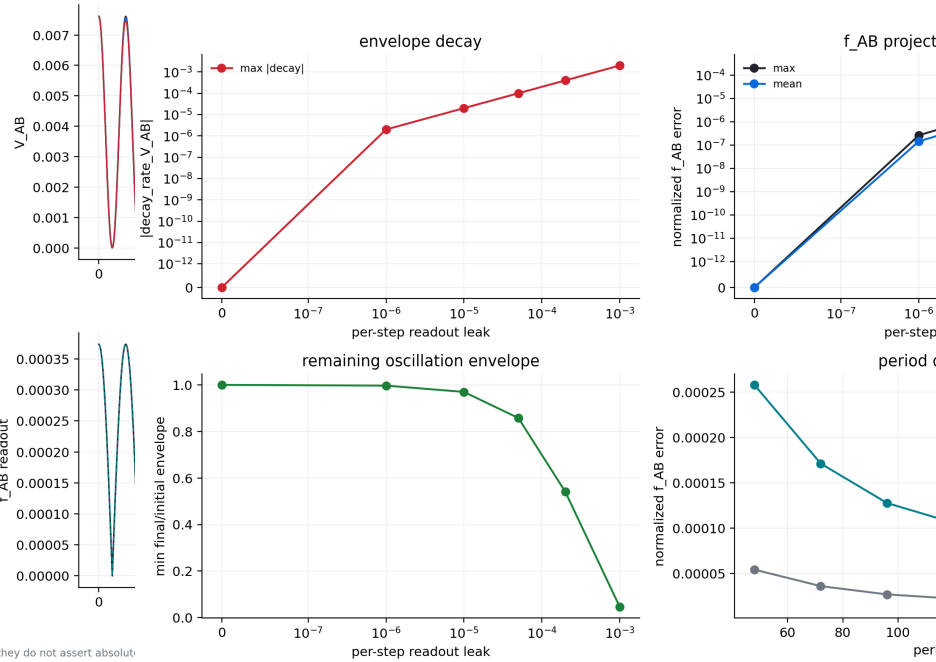
Main observation summary



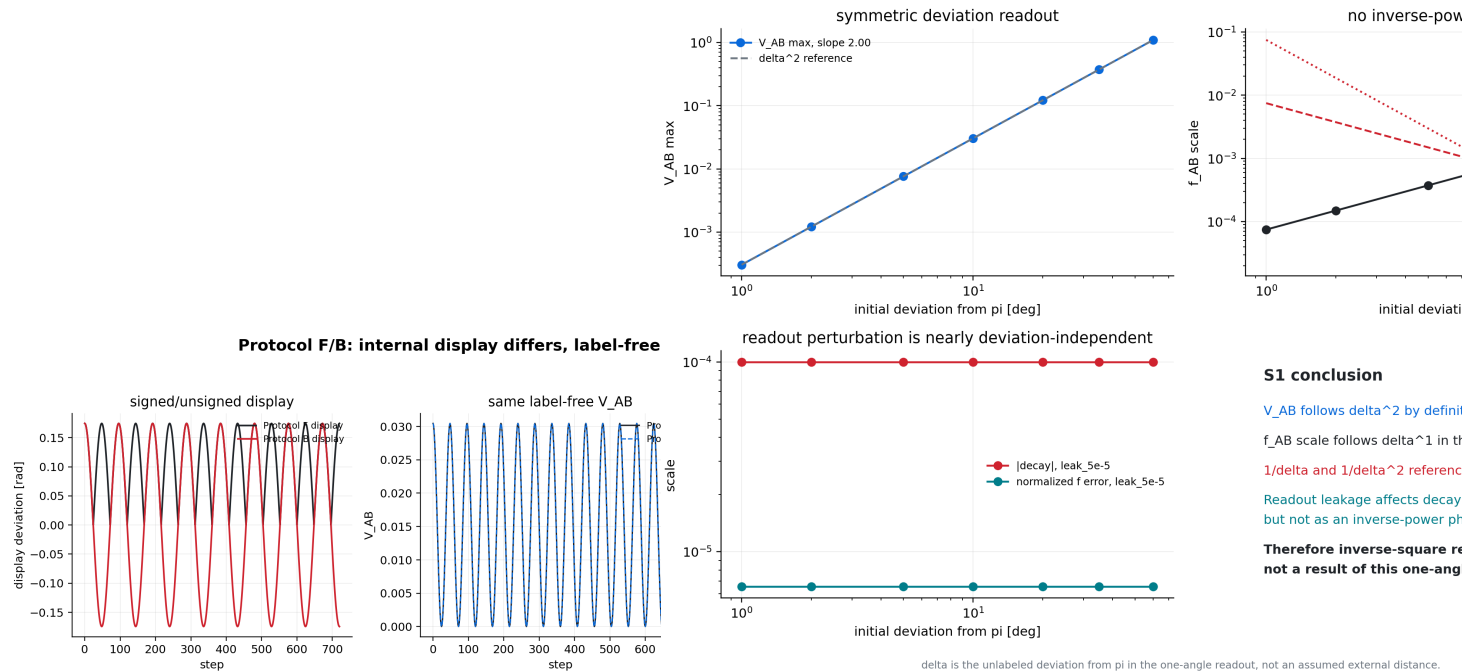
AB one-angle readout: harmonic state



Axes are readout coordinates for this figure only; they do not assert absolute

Readout wave response: damping and f_{AB} perturbati

Phase-difference scaling in the one-angle AB readout



S1 conclusion

V_{AB} follows δ^2 by definition

f_{AB} scale follows δ^1 in the

$1/\delta$ and $1/\delta^2$ reference

Readout leakage affects decay

but not as an inverse-power ph

Therefore inverse-square re

not a result of this one-ang

delta is the unlabeled deviation from pi in the one-angle readout, not an assumed external distance.

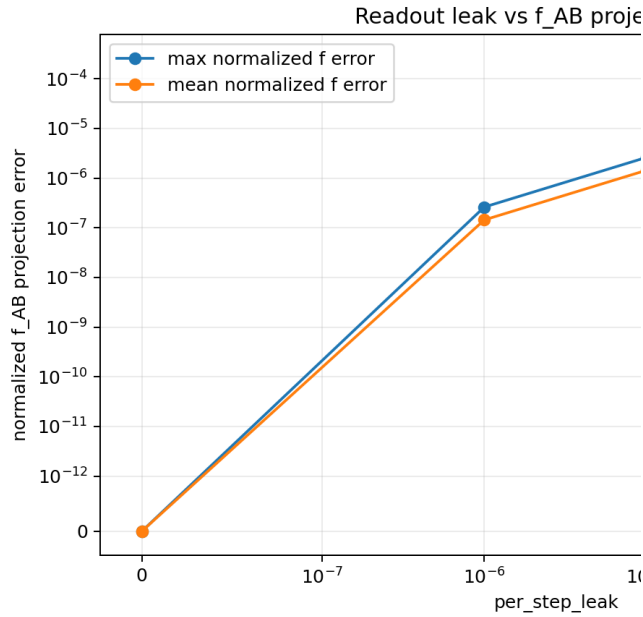
3. One-Angle Parameter Sweep

3.1 Main Result

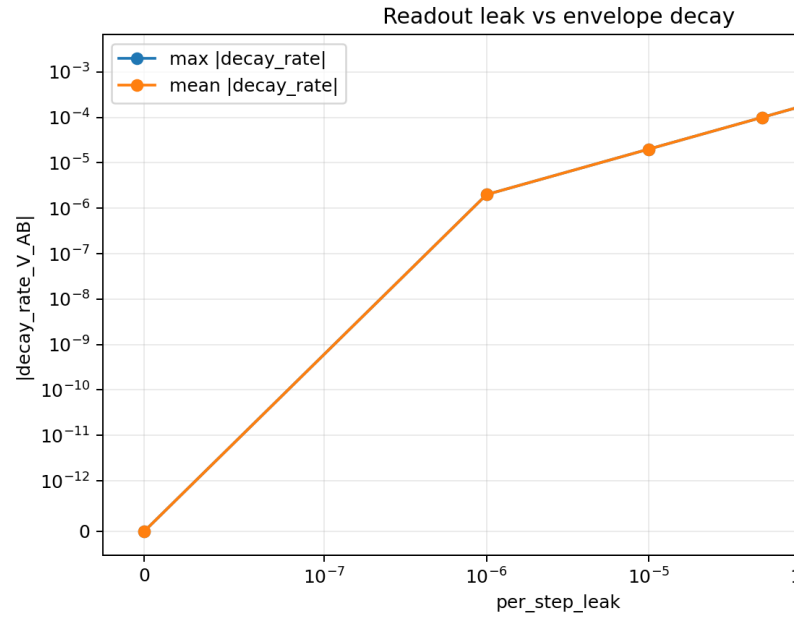
Quantity	Value
sweep_configuration_count	210
case_summary_count	420
period_count	5
deviation_count	7
leak_count	6
max_Q_closed_abs	0.0
label_free_protocol_degenerate_all_cases	True
oscillation_detected_all_cases	True
readout_off_decay_max_abs	8.26e-17
decay_abs_monotonic_by_leak_all_grids	True
max_normalized_f_AB_projection_error	2.58e-4

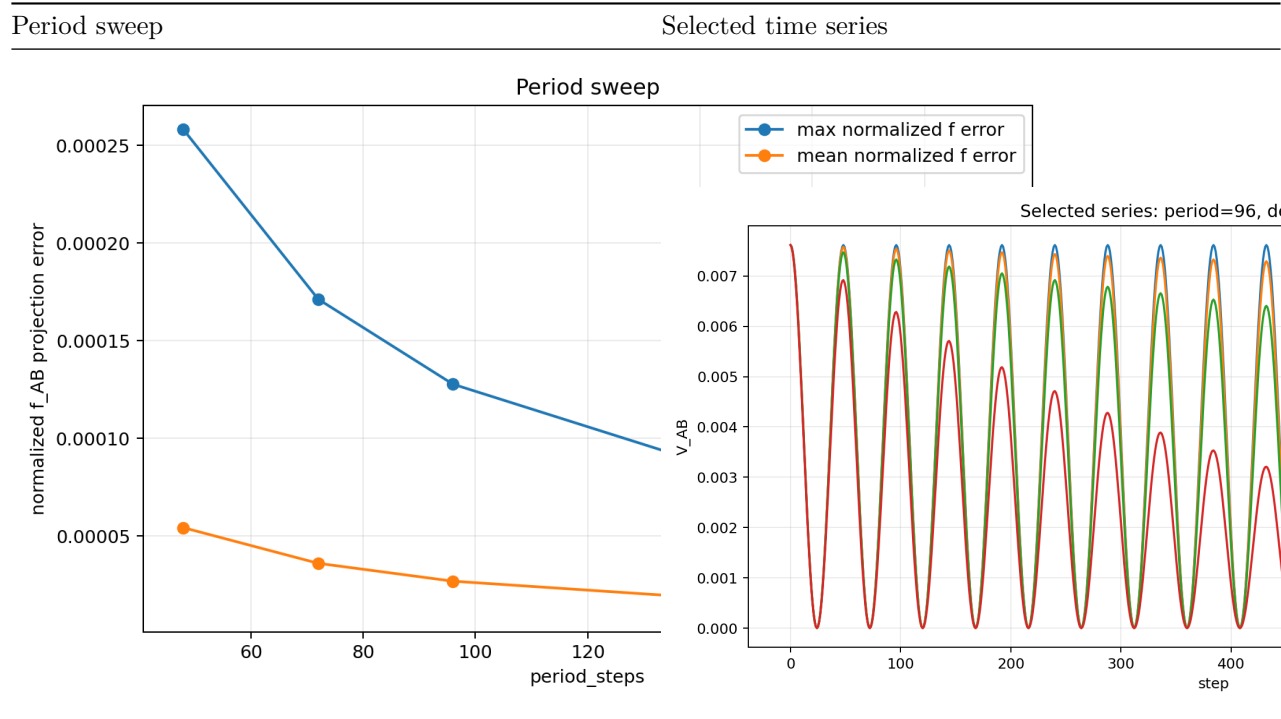
This sweep confirmed that the one-angle version is stable under weak readout, while a strong readout wave distorts the f_{AB} projection.

Leakage and projection error



Leakage and decay





4. c=1 Internal Calibration and chi-tau Area Sweep

4.1 Meaning of the Experiment

In the one-angle harmonic readout, distance attenuation according to phase difference was not read.

Therefore, the experiment step s was not treated as time itself. Instead, an independent temporal phase candidate

τ_{read}

was introduced.

The purpose was to test whether

χ_{read} and τ_{read} form an independent plane.

4.2 Main Result

Quantity	Value
case_summary_count	288
power_candidate_count	48
max_Q_closed_abs	0.0
disabled_max_area	0.0
locked_max_area	7.11e-15
independent_min_area	0.0024367633602631385
c1_readout_off_max_epsilon_c_abs	2.33e-15
c1_area_sweep_detected_all_cases	True
tau_is_step_used_any	False
external_c_used_any	False

The following was established.

tau disabled does not create area.

tau locked does not create area.

tau independent creates chi-tau area.

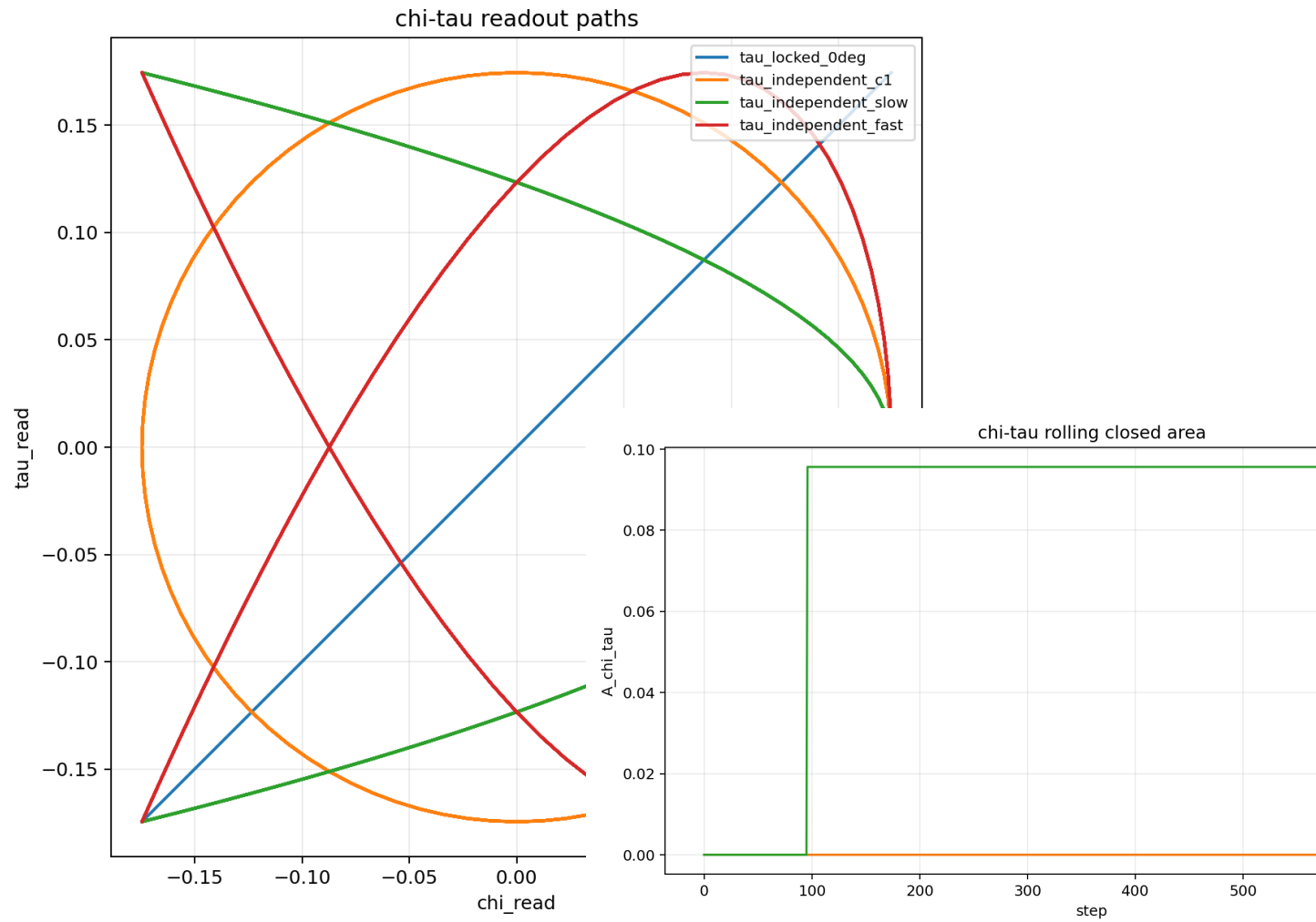
Thus, an independent temporal phase can make the chi-tau plane stand.

However, this alone does not generate distance attenuation.

4.3 Figures

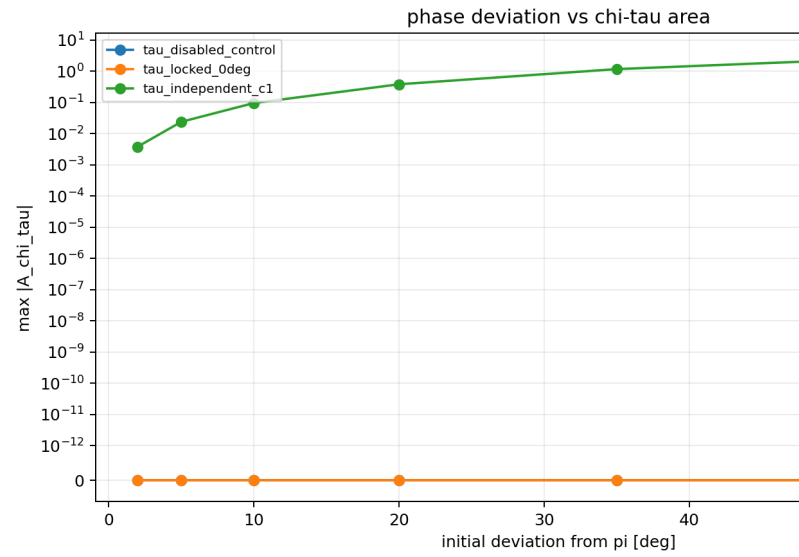
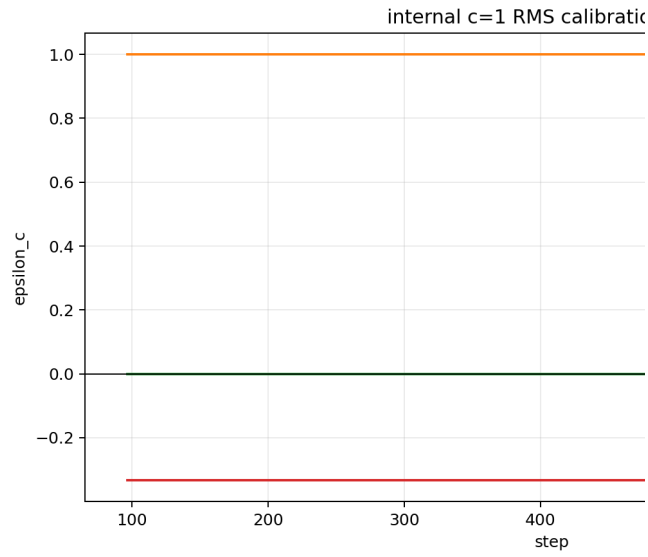
chi-tau surface

Area sweep

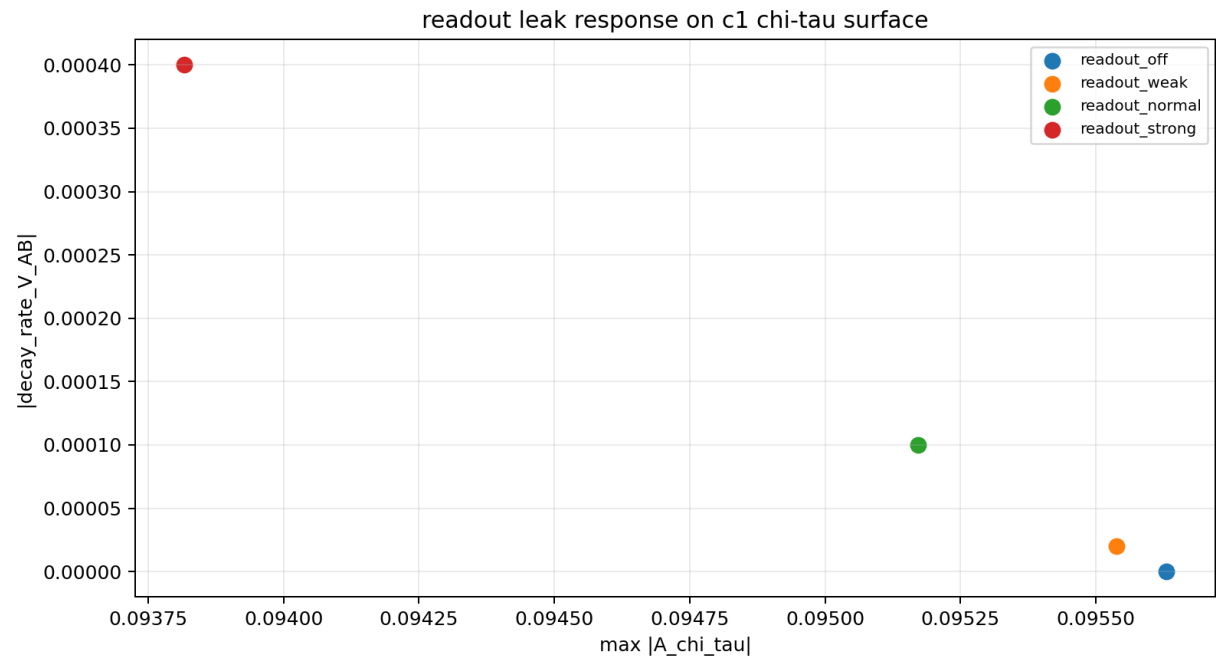


c=1 calibration error

Power candidates



Readout leakage



5. c=1 Internal Calibration Parameter Sweep

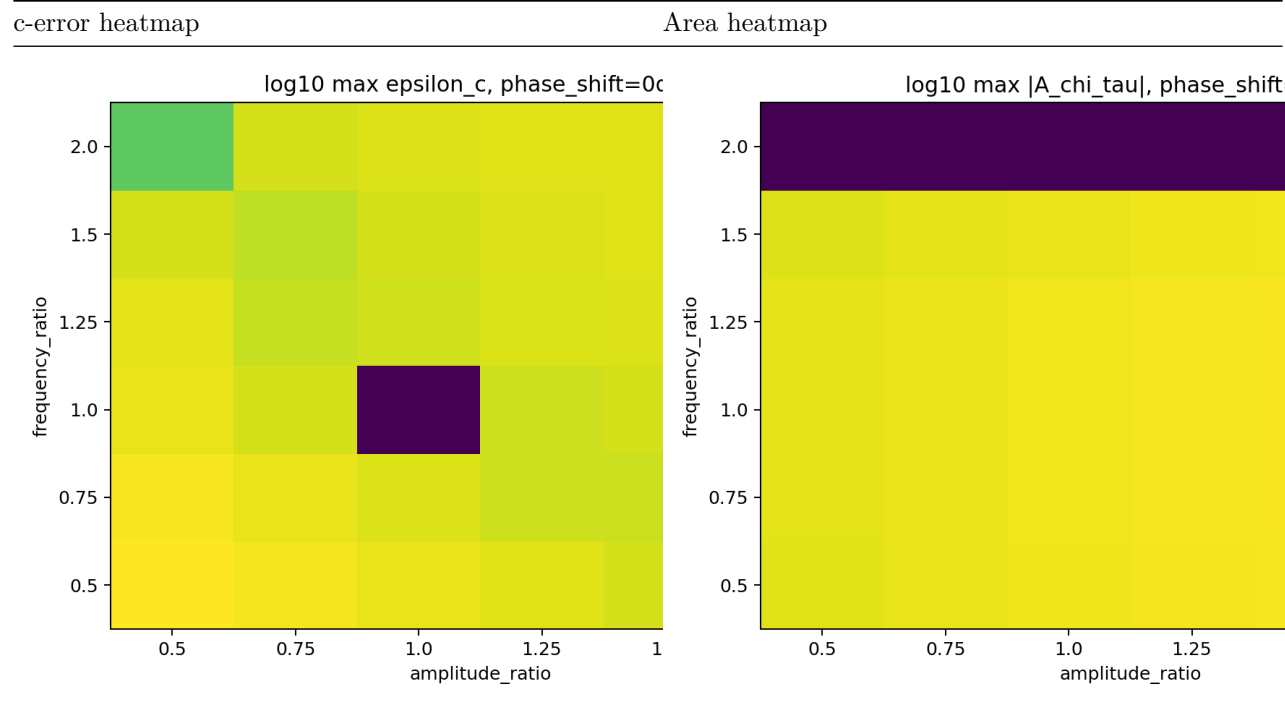
5.1 Main Result

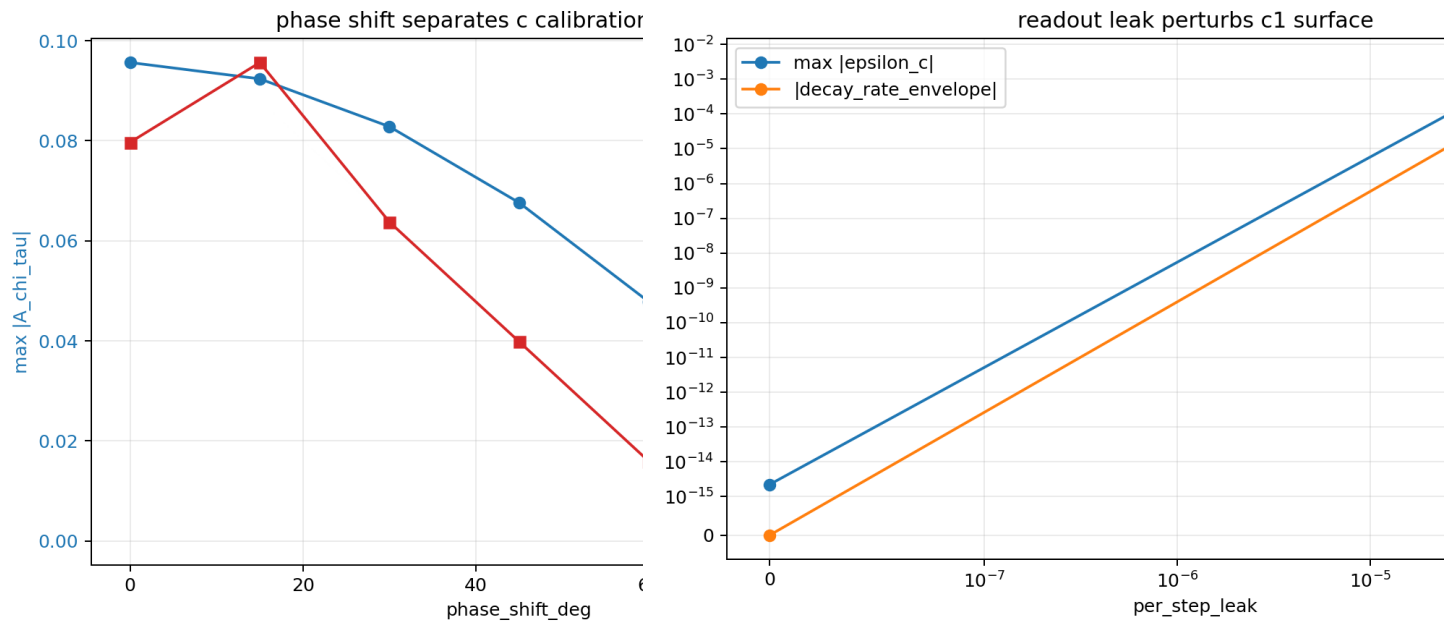
Quantity	Value
sweep_case_count	756
readout_off_case_count	252
rank2_readout_off_count	246
c1_surface_like_readout_off_count	13
c1_locked_like_readout_off_count	8
min_c_error_readout_off	1.11e-15
max_area_readout_off	0.19685536479742288

This clarified that the following three conditions must be required simultaneously:

```
c=1
rank_chi_tau = 2
A_chi_tau != 0
```

Being close to $c=1$ alone does not mean that a χ - τ surface has been formed.





6. Inverse-Area Compensation Diagnosis

6.1 Purpose

After the χ - τ area was formed, the experiment tested whether

$1 / A_{\chi\tau}$

would naturally remain on the closure-compensation side.

The important point is not to construct $1/A_{\chi\tau}$ as post-processing.

That is a constructed control and must be distinguished from native readout.

6.2 Main Result

Quantity	Value
diagnostic_case_count	288
area_valid_case_count	144
fit_count	576
native_fit_count	132
derived_fit_count	84
native_positive2_count	0
constructed_reciprocal_positive2_count	2

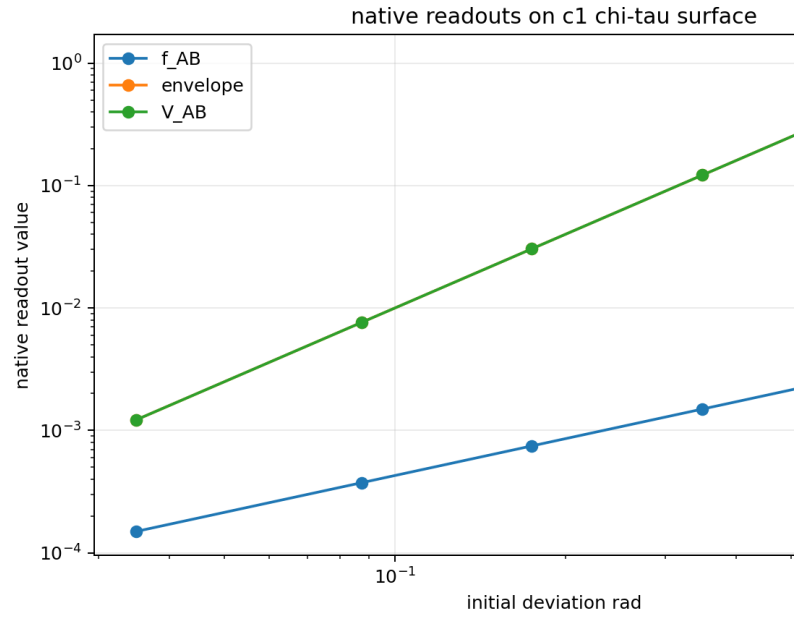
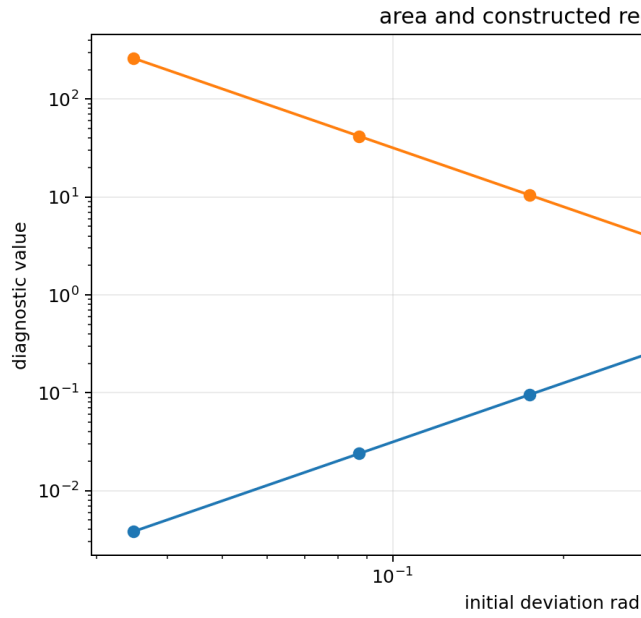
The result is clear.

If $1 / A_{\chi\tau}$ is constructed, $\alpha \approx 2$ appears.

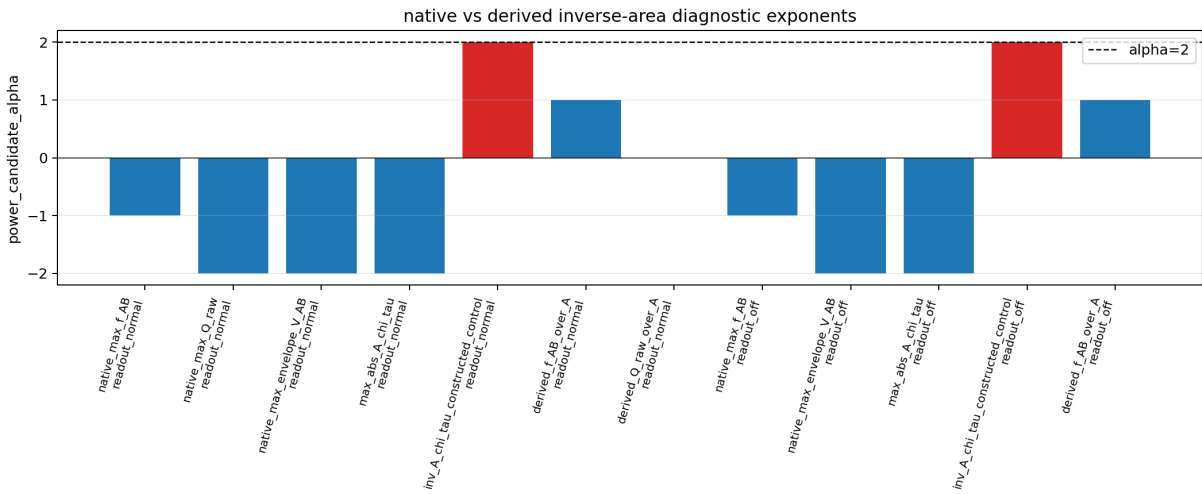
However, $\alpha \approx 2$ does not appear in native readout.

Constructed inverse-area control

Native candidates



Alpha comparison



7. Native Inverse-Area Extended Sweep

7.1 Main Result

Quantity	Value
sweep_case_count	1323
area_valid_case_count	1260
c1_surface_like_case_count	126


```

psi_A_in = (chi_pass + i eta_pass) / 2
psi_B_in = -(chi_pass + i eta_pass) / 2
Psi_out  = S_Delta Psi_in
chi_read = Re(psi_A_out - psi_B_out)
eta_read = Im(psi_A_out - psi_B_out)

```

The test is whether the acceleration-like harmonic readout from the AB two-body relation f_{AB} can still be evaluated in the same form when the recoil map is inserted as A/B two-channel scattering.

This section does not test:

```

AB asymmetrization
low-localization limit
instantaneous contraction
harmonic transfer
localization transfer

```

Those questions belong to a separate wave-packet focusing experiment line.

The present V4 evaluates the fermion-like recoil protocol within the current AB two-body acceleration-readout assumptions, in addition to the transmission-type central readout.

8.2 Recoil Map

Let the right-moving and left-moving incident channels be

```

\Psi^{\mathrm{in}}\}
=
\begin{pmatrix}
\psi_+^{\mathrm{in}}\} \\
\psi_-^{\mathrm{in}}\}
\end{pmatrix}.

```

Let Δ_F be the fermion-like exchange phase. Define the transmission and reflection amplitudes by

```

t_\Delta
=
e^{i\Delta_F/2}
\cos\frac{\Delta_F}{2}

r_\Delta
=
-i e^{i\Delta_F/2}
\sin\frac{\Delta_F}{2}.

```

The two-channel scattering matrix is

```

S_\Delta
=
\begin{pmatrix}
t_\Delta & r_\Delta \\
r_\Delta & t_\Delta
\end{pmatrix},

```

and the outgoing channels are read as

```

\Psi^{\mathrm{out}}\}
=
S_\Delta
\Psi^{\mathrm{in}}\}.

```

That is,

$$\begin{aligned}
& \psi_+^{\mathrm{out}} \\
& = \\
& t_{\Delta} \psi_+^{\mathrm{in}} \\
& + \\
& r_{\Delta} \psi_-^{\mathrm{in}} \\
& \psi_-^{\mathrm{out}} \\
& = \\
& r_{\Delta} \psi_+^{\mathrm{in}} \\
& + \\
& t_{\Delta} \psi_-^{\mathrm{in}}.
\end{aligned}$$

For the complete-recoil condition,

$$\Delta_F = \pi,$$

we obtain the equivalent condition

transmission amplitude $t_{\Delta} = 0$

reflection amplitude $r_{\Delta} = 1$

In this limit the outgoing channels are read as an exchange of the incident channels.

8.3 Pre-Experiment Criteria

The V4 additional protocol checks at least the following items.

Item	Criterion
Two-channel scattering	S_{Δ} acts on the A/B incident channels without using $q_{\mathrm{out_factor}}$ as an operator
Unitarity	Channel norm error remains within numerical error
Closure residual after recoil	Q_{closed} does not break down
Harmonic readout	The harmonic structure of D_{AB} , V_{AB} , and f_{AB} is maintained
Readout-leak response	The difference between readout-off and strong-readout conditions follows the same discipline as V1
Protocol comparison	Transmission, display-recoil, and fermion-like recoil protocols are all recorded
χ -tau surface	Independent τ_{read} area readout is not broken

8.4 Main Result

Quantity	Value
harmonic_case_count	48
c1_case_count	144
scattering_protocol_count	3
fermionic_delta_f	3.141592653589793
fermionic_reflection_rate	1.0
fermionic_transmission_rate	3.749399456654644e-33
fermionic_q_out_factor	-1.0
q_out_factor_applied_any	False
fermionic_full_two_channel_scattering_used_all_cases	True
fermionic_max_scattering_unitarity_error	4.163336342344337e-17
max_Q_closed_abs	0.0
fermionic_regular_cell_harmonic_consistent_nonstrong_modes	True

Quantity	Value
fermionic_strong_readout_perturbs_harmonic_projection	True
fermionic_max_f_AB_projection_error_regular_nonstrong	2.2756072064286935e-6
fermionic_max_f_AB_projection_error_regular_strong	9.072436576166898e-6
fermionic_reflection_event_cell_count_total	720
label_free_pass_vs_fermionic_match_all_cases	True
label_free_display_vs_fermionic_match_all_cases	True
readout_off_decay_max_abs	3.818134371020083e-18
readout_strong_decay_min_abs	4.0004000533409265e-4
fermionic_tau_disabled_max_area	0.0
fermionic_tau_locked_max_area	8.881784197001252e-16
fermionic_tau_independent_min_area	0.00375268307733001
fermionic_c1_readout_off_max_epsilon_c_abs	2.3314683517128287e-15
fermionic_c1_area_sweep_detected_all_cases	True
tau_is_step_used_any	False
external_c_used_any	False
f_A_or_f_B_used_any	False

The result is read as follows.

In the complete-recoil limit, $q_{\text{out_factor}}=-1$ is obtained as a diagnostic quantity. However, V4 does not multiply χ_{read} by $q_{\text{out_factor}}$ as an operator.

V4 acts with the two-channel scattering matrix on the A/B incident channels, and re-reads χ_{read} and η_{read} from the outgoing channel difference.

For all fermion-like recoil cases, $\text{full_two_channel_scattering_used}=\text{True}$, and the maximum unitarity error was $4.16\text{e-}17$.

Under that condition, the label-free readouts D_{AB} and V_{AB} matched the V1 transmission-type readout.

This is consistent with the Protocol F/B degeneracy confirmed in V1.

The display-only foldback protocol $\text{display_reflection}$ was also recorded as a control.

The display-recoil and fermion-like recoil protocols also matched under the label-free D_{AB} and V_{AB} readouts.

For harmonic readout, projection consistency was maintained under non- readout_strong conditions, as in V1.

Strong readout produced projection distortion, but this is the same readout-wave perturbation observed in V1 and is not a breakdown specific to the recoil map.

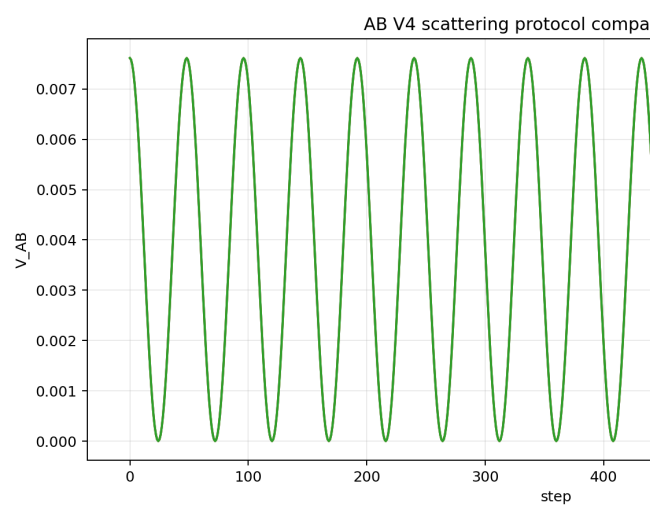
The χ - τ surface also maintained the following structure:

```
tau disabled: no area
tau locked: no area
tau independent c1: area present
```

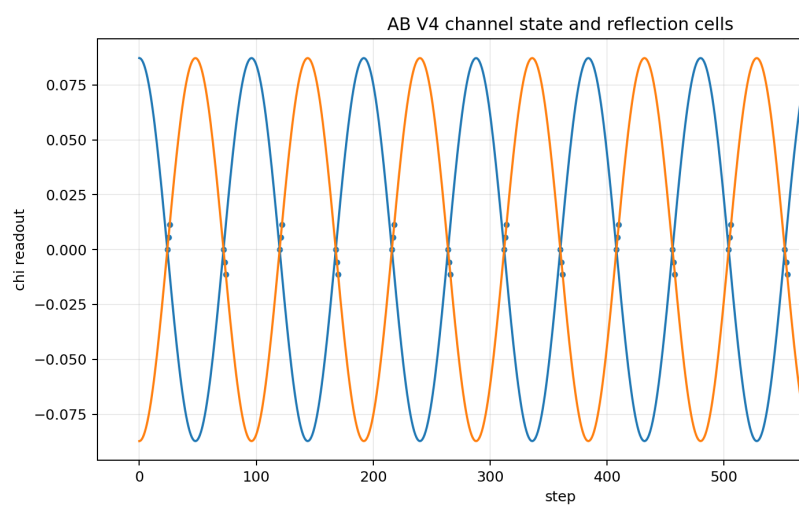
Therefore, even when the fermion-like recoil map is incorporated into AB two-body acceleration readout as A/B two-channel scattering, the acceleration-like harmonic readout and the χ - τ area readout are not broken within the present assumptions.

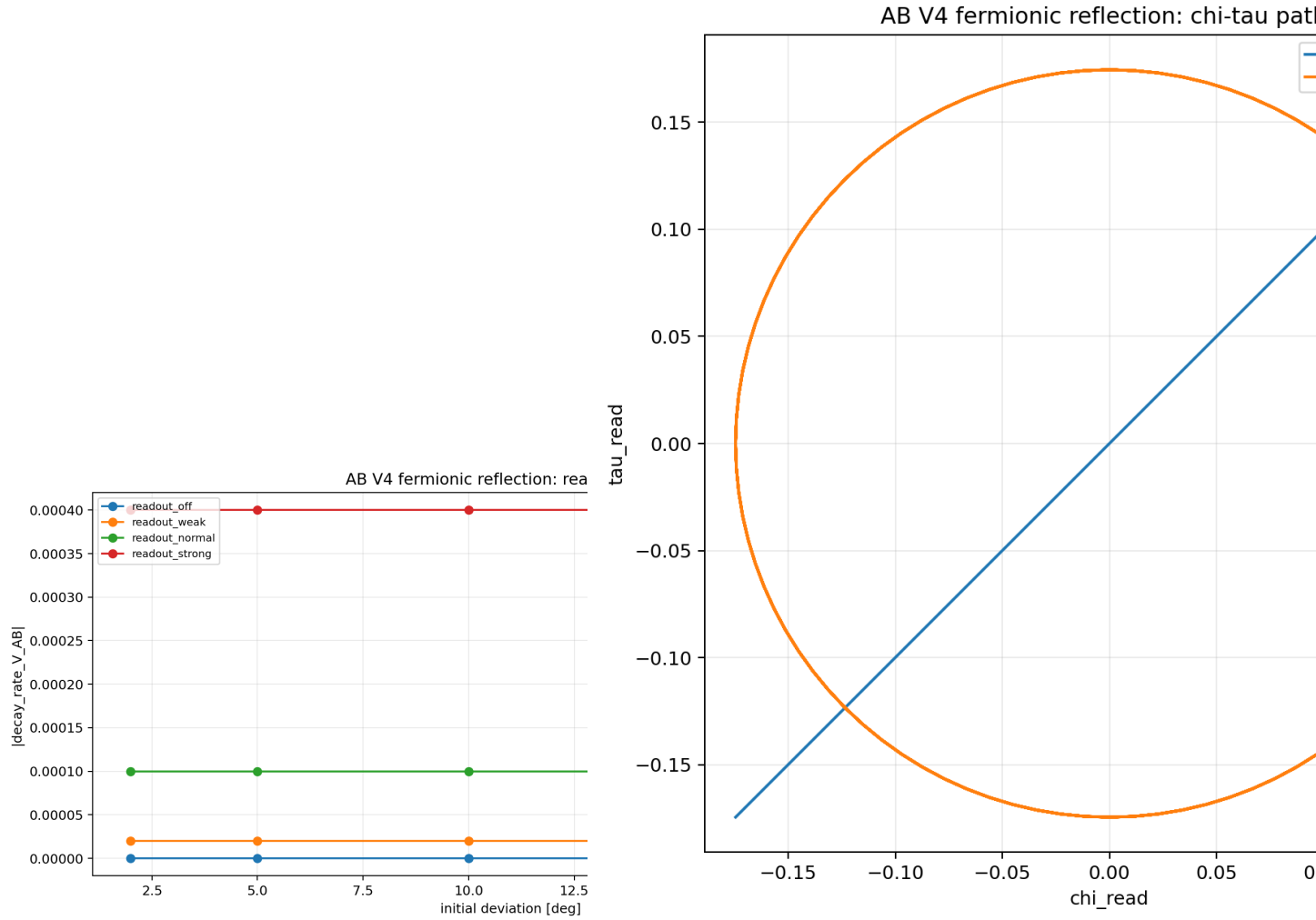
8.5 Figures

Protocol comparison



Channel state





9. Overall Judgment

9.1 Established

Item	Judgment
Read f_{AB} as a single AB relation	Established
Read harmonic oscillation without placing f_{A} or f_{B}	Established
Confirm label-free degeneracy of Protocol F/B	Established
Confirm disappearance of decay when the readout wave is stopped	Established
Read χ - τ area through independent τ_{read}	Established
Confirm $c=1$ as a candidate necessary condition	Established
Post-processed $1/A_{\chi\tau}$ shows $\alpha \approx 2$	Established
Apply the fermion-like recoil map as an A/B two-channel scattering matrix to the incident channels	Established

Item	Judgment
Do not use <code>q_out_factor</code> as an operator	Established
Keep the two-channel scattering unitarity error within numerical error	Established
Confirm that label-free <code>D_AB</code> and <code>V_AB</code> match <code>V1</code> even after introducing fermion-like recoil	Established
Maintain the independent- <code>tau_read</code> <code>chi-tau</code> surface even after introducing fermion-like recoil	Established

9.2 Not Established

Item	Judgment
Independent metrical determination of distance exponent in AB two-body system alone	Not established
Formation of temporal phase surface from <code>c=1</code> alone	Not established
Native inverse-law readout	Not detected
Native inverse-square readout	Not detected
Automatic conversion from <code>chi-tau</code> area to <code>1/A</code> compensation	Not detected
Correspondence to standard gravity	Not established

10. Interpretation

The main achievements of the AB two-body system are the following two points:

1. Using only the AB relation `f_AB`, an acceleration-like harmonic readout can be constructed.
2. The AB two-body system alone lacks an independent gauge for reading the distance exponent.

The first point is a positive result.

In the preceding ABC multigauge interference readout experiment, mass-like, momentum-like, and energy-like readouts were reconstructed from interference correlations.

The present experiment shows, as a next step, that an acceleration-like readout can be constructed from the harmonic AB two-body relation.

This is not the result of inserting a standard force from outside.

It is a harmonic displacement appearing as a closure readout of the two-body relation `f_AB`.

The second point is a boundary condition.

Inside the AB pair, A and B can be read only through one another.

Therefore, relative-distance change can be read, but there is no independent gauge that fixes whether the change rate should be measured as

proportional
inverse proportional
inverse square

This is not a failure of namelessness. Rather, it is a consequence of namelessness.

What cannot be read must not be declared read by post-processing.

In the `chi-tau` area sweep, an independent temporal phase allowed an area readout to stand.

However, this did not automatically provide a distance-exponent readout.

In addition, if $1/A_{\chi\tau}$ is constructed in post-processing, an inverse-square form naturally appears. But this is not native readout.

The judgment of this summary is therefore:

Acceleration-like readout was confirmed.

Distance-exponent readout cannot be determined in the AB two-body system alone.

The V4 additional protocol did not change this judgment.

In the complete-recoil limit, the diagnostic quantity $q_{\text{out_factor}}=-1$ was obtained, but V4 does not use it as an operator.

After the A/B two-channel scattering matrix acted on the incident channels, the label-free readouts D_{AB} and V_{AB} matched the V1 transmission-type readout.

This means that the V1 result, in which recoil-type and transmission-type protocols are degenerate under AB two-body nameless readout, was confirmed again using the two-channel scattering-matrix recoil map.

11. Connection to the Next Experiment

The next stage following this summary is the ABC three-body system.

However, the third wave C is not an external absolute observer.

C is itself part of the closed phase system and generates

f_{AC}
 f_{BC}
 f_{ABC}

Therefore, the ABC experiment must test:

whether C can be used as an independent metrical gauge,
whether C contaminates the main AB readout,
whether f_{ABC} can be used as representative time,
and whether f_{AB} , f_{AC} , and f_{BC} can be separated as circumferential-direction candidates.

References

Self-References

1. Noriaki Kihara, "Basic Axiom System of the Nameless Equal-Amplitude Composite Wave Model v4", Version DOI: [10.5281/zenodo.21316620](https://doi.org/10.5281/zenodo.21316620), Concept DOI: [10.5281/zenodo.21315735](https://doi.org/10.5281/zenodo.21315735), 2026.
2. Noriaki Kihara, "Construction Experiment of Multigauge Interference Readout Conserved Quantities in an ABC Closed Phase System", Version DOI: [10.5281/zenodo.21332875](https://doi.org/10.5281/zenodo.21332875), Concept DOI: [10.5281/zenodo.21308049](https://doi.org/10.5281/zenodo.21308049), 2026.
3. Noriaki Kihara, Definitional Supplement on Label-Free Two-Arc Relative Phase and Harmonic Readout in an AB Two-Body Closed Phase System, 2026.
4. Noriaki Kihara, Experiment Specification for One-Angle Circumferential Phase Harmonic Readout in an AB Two-Body Closed Phase System v1, 2026.
5. Noriaki Kihara, Experiment Specification for $c=1$ Internal Calibration and Spatial-Phase/Temporal-Phase Area Sweep Readout in an AB Two-Body Closed Phase System v1, 2026.
6. Noriaki Kihara, Definitional Supplement on Internal Closure of Self-Terms and Separation of N-Body External Readout in Closed Complex Phase Waves, 2026.

External References

The external references are used only as minimal standard background on wave readout, phase, and observational probability, not as derivational premises of this paper.

7. Max Born, "Zur Quantenmechanik der Stossvorgaenge", *Zeitschrift fuer Physik* 37, 863-867, 1926. DOI: **10.1007/BF01397477**.
8. Y. Aharonov and D. Bohm, "Significance of electromagnetic potentials in the quantum theory", *Physical Review* 115, 485-491, 1959. DOI: **10.1103/PhysRev.115.485**.